SAMPLE

Magnetism

Fields, Forces, and the Magnetic Universe

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Introduction

A magnet is one of the first scientific instruments a child ever holds. Two colored bars that refuse to touch, or snap together with a click you can feel through your fingers, teach the oldest lesson in physics: the world pushes and pulls without touching. Long before anyone could explain it, the magnet was evidence that invisible forces are real, patient, and lawful.

This book is about that force. Magnetism turns out to be one of the deepest subjects in physics, and one of the most personal. Every compass reading, every refrigerator note, every hospital scanner, every spinning motor, every recorded song on a hard drive rides on it. The modern world is welded together by magnets in ways most people never see, and the universe beyond the world is even more magnetic than the world itself. The Sun is a magnet the size of three hundred thousand Earths. Dead stars squeeze magnetic fields until they become engines of radiation. The space between the stars is threaded by fields older than the planet.

The story runs from a strange black stone found near a Greek town to the edge of modern physics, and it keeps its honesty along the way. The ancient world knew the lodestone attracted iron and could not say why. Medieval sailors trusted the compass with their lives while scholars argued about whether the pole star itself was the magnet. In 1820 a Danish philosopher watched a compass needle twitch beside a wire and cracked the whole subject open. In 1831 an English bookbinder's apprentice showed that magnetism could be made to pour electricity from motion, and the electrical age began. In the twentieth century the theory of the atom explained why iron clings and wood does not, and the twenty-first added new families of magnetic materials whose names, altermagnetic among them, are still settling into the textbooks.

The plan of the book is simple. It begins with the discovery: the lodestone, the legend of the shepherd, the Chinese spoon that pointed south, the medieval letter that first described the pole, and the Elizabethan physician who turned a curiosity into a science. Then it builds the grammar of the subject: what a field is, what poles are, why no one has ever found a magnet with one end, and how the atom itself came to be understood as a tiny magnet. It walks through the family of magnetic materials, from the strong and familiar to the feeble and strange, and explains honestly why most things are called non-magnetic when in truth nothing is.

From there the book turns to what magnetism does. The Earth is a magnet whose stuttering pole guides navigators and guards the atmosphere. The dynamo and the motor built the industrial world. The medical scanner reads the body's own magnetism. And beyond the Earth, the largest magnetic structures in existence braid and snap, hurling storms that have burned telegraph offices and darkened cities.

Two promises hold throughout. First, no equations: every idea is carried in plain language, pictures, and honest numbers, and where a number is rounded or a claim is still open, the text says so. Second, claims are graded. Ancient legends are labeled as legends, folk remedies as mostly placebo, frontier physics as promising but unsettled. The magnets on your refrigerator will still hold the shopping list tomorrow; the purpose of this book is to show how much more they know than they let on.

Chapter 1: The Stone That Pointed

No one knows who first picked up a lodestone and felt the tug. The safest honest statement is that somewhere in the ancient world, probably in more than one place, a person noticed that a certain heavy black stone pulled at iron nails, and that the stone, hung on a thread, swung around to face the same direction every time. That single doubled observation, attraction and alignment, is the seed of the entire subject.

The stone is magnetite, an iron oxide, one of the few minerals that is naturally magnetized strongly enough to act as a magnet as it comes out of the ground. The pieces that do so are called lodestones, and they are rare: only a small fraction of magnetite is found magnetized, and a large fraction of that has been gathered by collectors over the centuries. The name carries its own history. The ancient Greeks called it the Magnesian stone, after the district of Magnesia in Thessaly where good specimens were found, and the modern word magnet descends straight from that place name.

The legend is better known than the fact. A shepherd, the story goes, wandering Mount Ida with his flocks, found the nails pulled from his sandals and the tip of his staff clinging to the ground. The tale appears in a Roman poem of the first century and has been retold in a dozen forms since, sometimes with different mountains and different herders. It deserves its label as a legend: no location, date, or name survives, and the details have the smell of an invention built backwards from a familiar stone. What is genuinely ancient is the observation itself. Aristotle, writing in the fourth century BCE, credits Thales of Miletus, a century earlier, with the first reasoned discussion of the stone's power. Thales reportedly thought it possessed a soul, because it moved iron, which for a philosopher of his era was a serious claim about what counts as alive.

China gives the story a second beginning, and it is the more practical one. A Chinese text from the fourth century BCE describes the magnet's attraction as if the stone were summoning its children. By the first century BCE, Chinese writers describe the south-pointing spoon: a lodestone carved into the shape of a ladle, set on a polished bronze plate, its handle swinging to the south. The instrument was ceremonial and imperfect, and its role in actual navigation is debated, but the principle it demonstrated, that a free magnet aligns with the Earth, is the principle the compass still uses. Historians label the Chinese claims of a south-pointing chariot separately: that device, an earlier invention, worked by gears and had nothing to do with magnetism, and texts sometimes blur the two.

The Mediterranean world noticed the alignment trick later and more slowly. Roman authors describe shepherds and sailors using magnetite, but the compass as a navigation tool enters European writing clearly only around the twelfth century, and its first European mention, a poem of about 1190, treats it as a sailors' secret already well known at sea. Meanwhile Arabic authors, including writers in the twelfth century who passed the device between the two worlds, described the floating needle: a sliver of touched iron set on a straw or a cork riding water, turning until it found north. The needle had to be magnetized by stroking it against a lodestone, which raised a question that would not be properly answered for six hundred years: why does iron take magnetism from a stone, and keep it?

Medieval Europe produced the first experimental study of the magnet, and it is a strikingly modern document. In 1269 a crusader camped at the siege of Lucera, Pierre de Maricourt, known as Peregrinus, wrote a letter to a friend laying out systematic tests. He shaped lodestones into spheres, mapped their directions with a needle, found the two points where the pull ran straight down into the stone, and named them poles. He demonstrated that like poles repel and unlike attract, showed that breaking a lodestone yields two complete magnets rather than a separated north and south, and even sketched a perpetual-motion machine driven by magnets, which was wrong, and honestly wrong in an instructive way: the error forced later workers to ask why the magnets would not spin, and the answer deepened the theory. The letter is the first European use of the word pole for a magnet.

The stone itself deserves a closer look before the story moves on, because its rarity is part of the wonder. Magnetite is common enough as an iron ore, mined for steel on every continent, but only a tiny fraction of it arrives from the ground already magnetized, and no one can point to a lodestone deposit the way one points to a gold field. Some specimens carry enough magnetism to lift many times their own weight in iron, and collectors still prize a good lodestone the way earlier centuries did: a museum-grade specimen is a working instrument, not a mineral curiosity. The deep origin of that magnetization is itself a graded question this book returns to, since the leading explanation involves lightning striking ore near the surface and leaving the field of the strike frozen in the rock.

The chapter's last figure waits in Elizabethan London, but he belongs to the next chapter, because he changed the question itself. For two thousand years the puzzle had been the stone. William Gilbert asked what the stone was pointing at, and the answer turned the lodestone from a curiosity into the key to the Earth.